

THE EXHAUSTION OF ONE-DIMENSIONAL PROPAGATION: A STRUCTURAL NO-GO FOR THE RIEMANN HYPOTHESIS, AND TWO DIRECTIONS THAT SURVIVE IT

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ABSTRACT. A research programme spanning thirty-five phases has reduced the Riemann Hypothesis (RH), under the Connes–Consani–Moscovici (CCM) framework, to the vanishing of a single non-negative measure $d\nu = dm_{\text{full}} - dm_{\text{full, on}}$, and has catalogued seven structural walls (MW-1 through MW-7) that block every attempted proof. We make two contributions. *First (the no-go)*: we isolate the common logical form of all seven walls. Every live or recently closed route presents RH as the *boundary point* of a one-parameter family whose every other member is an unconditional theorem (the De Bruijn–Newman time t , the spectral-triple approximation parameter x , the place-set S , the Abel scale λ). We argue that the walls are precisely the statement that *no one-dimensional propagation principle reaches that boundary*, and we verify this by exhibiting, for each classical 1D mechanism — limit/continuity, analyticity, parabolic backward uniqueness, monotonicity/integrality, ergodic recurrence — the exact point at which it either is excluded by a wall or coincides with RH itself. As a sharpening internal to the framework we prove that the Abel kernel W_λ is *strictly positive at every real point*, whence the universal quantifier in the trace criterion “ $T_\lambda = 0$ for all λ ” is redundant: a single scale suffices, and the λ -family collapses to a point. *Second (the survivors)*: we show the no-go leaves exactly two logical forms unexcluded — *two-dimensional propagation* (coupling the heat time with the arithmetic filtration via a maximum or monotonicity principle on a quadrant) and *index amplification* (a non-grouplike tensor operation on the Pontryagin space (K, Q) realising the Deligne mechanism: integer invariant, multiplicative defect, sublinear uniform bound). We give each a precise functional specification, identify what would falsify it, and state the open problems. We do not solve them.

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1. INTRODUCTION

1.1. What this paper is, and is not. This is not a paper that proves the Riemann Hypothesis, nor one that claims partial progress toward a proof. It is a paper about the *shape* of the obstruction. After a long programme of attempts—each documented, each refuted or reduced to a known wall—we are in the unusual position of possessing a near-complete map of why RH resists the methods currently available. A complete map of failure is itself a mathematical object, and it has a use: it constrains, by elimination, the logical form that any successful argument must take. The purpose of this paper is to extract that constraint rigorously and to deduce from it where the search should go next.

We work throughout inside the CCM framework [1], in the form developed in [6, 7]. We recall the minimum needed in §2. The two halves of the paper are independent in spirit: §3–§5 are a structural no-go argument (with one new unconditional theorem, §4); §6–§8 are a forward research agenda with falsifiable specifications but no theorems claiming to reach RH.

1.2. The one new theorem. The only genuinely new unconditional result proved here is Theorem 4.2: the Abel kernel W_λ of the CCM trace criterion is strictly positive at every real point (not merely almost everywhere), and consequently the trace criterion of [6] holds at a *single* scale λ rather than requiring all $\lambda > 0$. This is modest in difficulty—a three-term-recurrence argument—but it is exactly the kind of statement the structural analysis predicts should be available: it removes a spurious degree of freedom from the map without touching the hard core.

2. THE FRAMEWORK, RECALLED

We fix notation following [6, 7].

- The archimedean spectral measure is $dm_\infty(s) = (2\pi)^{-2} |\Gamma(1/4 + is/2)|^2 ds$ on $\mathbb{R}$, a probability measure (after normalisation) with density decaying like $|s|^{-1/2} e^{-\pi|s|/2}$.
- $\{\mathcal{P}_n\}_{n \geq 0}$ are the orthonormal polynomials for dm_∞ , with $\mathcal{P}_0 \equiv 1$ and Jacobi data $b_n = 0$ (parity), $a_n^\infty = \frac{1}{2} \sqrt{(2n+1)(2n+2)}$. They obey the three-term recurrence

$$(1) \quad s \mathcal{P}_n(s) = a_n^\infty \mathcal{P}_{n+1}(s) + a_{n-1}^\infty \mathcal{P}_{n-1}(s), \quad a_k^\infty > 0.$$

- $dm_{\text{full}} = |\zeta(1/2 + is)|^2 dm_\infty$ and $dm_{\text{full, on}} = |\zeta_{\text{on}}(1/2 + is)|^2 dm_\infty$, where ζ_{on} is the Hadamard product of ζ restricted to its on-line zeros.
- The central object is the difference measure $d\nu = dm_{\text{full}} - dm_{\text{full, on}}$. By [7,] it is *unconditionally a non-negative measure*: $d\nu = (R - 1) |\zeta_{\text{on}}|^2 dm_\infty$ with $R = |\zeta/\zeta_{\text{on}}|^2 \geq 1$, each off-critical zero quadruple contributing a factor ≥ 1 .
- The Abel kernel is, for $N = N(\lambda) := \#\{n : \gamma_n \leq \lambda\}$,

$$(2) \quad W_\lambda(s) = \sum_{n=1}^N n |\mathcal{P}_n(s)|^2 + (a_N^\infty)^2 |\mathcal{P}_{N+1}(s)|^2 \geq 0.$$

- The trace is $T_\lambda = \int_{\mathbb{R}} W_\lambda d\nu$. The CCM trace criterion [6, Thm] states

$$(3) \quad \text{RH} \iff T_\lambda = 0 \text{ for all } \lambda > 0.$$

- [7,] lists eight equivalent reformulations of RH as functionals of $d\nu$ ($d\nu = 0$; the Cauchy transform $\mathcal{G} \equiv 0$; $T_\lambda = 0$; the Jensen defect $\delta = 0$; the variational distance $\tilde{\delta}_\lambda^2 = 0$; $\text{KL} = 0$; $d_{\text{CCM}} = 0$).

Definition 2.1 (The Hadamard barrier, MW-7). No property of ζ that is verifiable without knowledge of the positions of its zeros implies $d\nu = 0$.

The Hadamard barrier is the framework-internal face of the master obstruction. Its companions across the programme are: MW-1 (Weil positivity $\equiv$ RH), MW-2 (the arithmetic propagation wall: the Euler product converges only for $\text{Re}(s) > 1$ and its information does not propagate to the critical strip; zeros are global objects), MW-3 (no Hasse–Minkowski in infinite dimension), MW-4 (every unconditional tool yields a positivity bound of the wrong sign), MW-5 (the arithmetic-site factorisation requires the unfinished Connes–Consani programme), MW-6 (uniform spectral-gap control $\equiv$ RH). The thesis of §3–§5 is that these seven are seven faces of one fact.

3. THE META-PATTERN: RH AS A BOUNDARY POINT

3.1. The recurring shape. Across the programme, every route to RH—whether eventually closed or still live—instantiates one template. There is a one-parameter family of mathematical objects; for all parameter values except one boundary value the relevant statement is an unconditional theorem; and the boundary value is exactly RH. We tabulate the instances, with their sources.

Family	Parameter	Unconditional control	Boundary $\iff$ RH
Self-approximation SR_d	d	proved for all $d \neq 0$	$d = 0$
DBN flow $T_\lambda(t)$	t	$T_\lambda(t) = 0$ for $t \geq \Lambda$	$t = 0$
Spectral triples H_x	x	real spectrum, all finite x	$x = \infty$
Semilocal index μ_S	S	positivity for small S	$S = \text{all places}$
Pontryagin $\kappa_\lambda(N)$	N	controllable, all finite N	$N = \infty$
Euler product	σ	converges for $\sigma > 1$	$\sigma = 1/2$

The sources are, respectively: the Bagchi/Voronin self-approximation family (proved for all $d \neq 0$ by Baker, Nakamura, Pańkowski, Garunkštis; the closing of $d = 0$ is RH [7,]); the De Bruijn–Newman flow with Λ the DBN constant [7, Docs 70–71]; the CCM finite-Euler spectral triples [2] analysed in [7,]; the semilocal Weil positivity of Connes–Consani [3] analysed in [7,]; the Pontryagin negative-index packet [7,]; and the classical Euler product itself.

Meta-principle 3.1 (The boundary-point pattern). In every analysed approach, RH is the limit/boundary member of a one-parameter family whose interior is unconditionally controlled. The structural walls MW-1–MW-7 are, jointly, the assertion that no one-dimensional propagation principle carries the interior control to the boundary.

We do not claim Meta-principle 3.1 as a theorem; it is an empirical regularity of the corpus, stated to be tested. §5 gives it teeth by checking the propagation mechanisms one by one.

4. SHARPENING: THE ABEL FAMILY COLLAPSES TO A POINT

Before the no-go we record the one place where the boundary-point pattern degenerates favourably: the Abel scale λ is not a genuine parameter at all.

Lemma 4.1 (No shared zeros of consecutive orthogonal polynomials). *For every $n \geq 0$, the polynomials $\mathcal{P}_n$ and $\mathcal{P}_{n+1}$ have no common zero in $\mathbb{C}$.*

Proof. Suppose $\mathcal{P}_{n+1}(s_0) = \mathcal{P}_n(s_0) = 0$. Equation (1) at index n gives $0 = a_n^\infty \mathcal{P}_{n+1}(s_0) + a_{n-1}^\infty \mathcal{P}_{n-1}(s_0) = a_{n-1}^\infty \mathcal{P}_{n-1}(s_0)$, so $\mathcal{P}_{n-1}(s_0) = 0$ since $a_{n-1}^\infty > 0$. Descending induction yields $\mathcal{P}_0(s_0) = 0$, contradicting $\mathcal{P}_0 \equiv 1$. $\square$

Theorem 4.2 (Strict pointwise positivity of W_λ). *For every λ with $N(\lambda) \geq 1$, $W_\lambda(s) > 0$ for all $s \in \mathbb{R}$.*

Proof. By (2), W_λ is a sum of non-negative terms with strictly positive coefficients. If $W_\lambda(s_0) = 0$ then every term vanishes; in particular the term $n = N$ (present because $N \geq 1$) forces $\mathcal{P}_N(s_0) = 0$, and the last term forces $\mathcal{P}_{N+1}(s_0) = 0$. This contradicts Lemma 4.1. $\square$

Remark 4.3. This strengthens [7, Cor. 7.5], which established $W_\lambda > 0$ off a finite set (the nodes of $\mathcal{P}_{N+1}$) and neutralised that set using the absolute continuity of $d\nu$. The exceptional set is in fact empty, and the conclusion needs no hypothesis on $d\nu$. For the degenerate range $N(\lambda) = 0$ one has $W_\lambda = (a_0^\infty)^2 |\mathcal{P}_1|^2$, vanishing only at $s = 0$, and the original argument applies.

Theorem 4.4 (A single scale suffices). *Fix any $\lambda_0 > 0$ with $N(\lambda_0) \geq 1$. Then $\text{RH} \iff T_\lambda[\lambda_0] = 0$. The universal quantifier in (3) is redundant.*

Proof. ($\Rightarrow$) Under RH, $d\nu = 0$, so $T_\lambda[\lambda_0] = 0$. ($\Leftarrow$) Write $T_\lambda[\lambda_0] = \int W_\lambda[\lambda_0] (R - 1) |\zeta_{\text{on}}|^2 dm_\infty$ with $R - 1 \geq 0$ ([7,]) and $W_\lambda[\lambda_0] > 0$ everywhere (Theorem 4.2). The integrand is non-negative; its integral vanishing forces $(R - 1) |\zeta_{\text{on}}|^2 = 0$ dm_∞ -a.e.; since $|\zeta_{\text{on}}|^2 > 0$ off a null set, $R \equiv 1$ a.e., i.e. $d\nu = 0$, i.e. RH ([7,], condition 2). $\square$

The Abel family is thus not a propagation problem at all: its boundary is already glued to its interior. This is the exception that proves the rule. For every *other* family in the table, the boundary is genuinely detached, and §5 explains why no 1D principle can reattach it.

5. THE NO-GO: ONE-DIMENSIONAL PROPAGATION IS EXHAUSTED

We now run through the classical mechanisms by which an unconditional interior statement could be propagated to a boundary value, and show each is either excluded by a named wall or is logically equivalent to RH. The list is intended to be exhaustive over the 1D toolkit; we invite the reader to supply a mechanism not reducible to one of these entries.

5.1. Limit / continuity. If the family $f(\theta)$ were continuous at the boundary and identically zero on the interior, the boundary value would vanish by continuity. For $H_x \rightarrow H_\infty$ this is exactly the route: Hurwitz’s theorem applied to the regularised determinants would transfer reality of zeros from finite x to $x = \infty$ [7, Prop. 5.3]. But the quantitative core of that limit is $\lim \mu_\lambda \geq 0$, the semilocal Weil positivity in the limit—this is MW-1 wearing the mask of MW-6. Continuity to the boundary *is* the prize, not a tool for it.

5.2. Analyticity. This deserves a clean statement because it explains a recurring frustration.

Proposition 5.1 (No faithful analytic detector). *Suppose $F : [0, \infty) \rightarrow \mathbb{R}$ is real-analytic and satisfies $F(t) = 0$ if and only if the zeros of Ξ_t are all real. Then RH holds.*

Proof. The DBN flow has Ξ_t with only real zeros for all $t \geq \Lambda$ (definition of the DBN constant Λ , with $\Lambda \geq 0$ known unconditionally and $\Lambda \leq 0$ being RH). Hence $F \equiv 0$ on $[\Lambda, \infty)$. A real-analytic function vanishing on a set with a limit point vanishes identically, so $F \equiv 0$, whence $F(0) = 0$, i.e. the zeros of $\Xi_0 = \Xi$ are all real: RH. $\square$

Proposition 5.1 shows that constructing a real-analytic faithful detector of zero-reality is *already* a proof of RH. This is why the natural detector $T_\lambda(t)$ is non-analytic precisely at $t = \Lambda$: the on/off split that defines $dm_{\text{full, on}}$ jumps there. Analyticity is not an available tool; it is the trophy. Any candidate “analytic monotone quantity” must be checked against this proposition—if it is faithful and analytic, the work is already done, which means it cannot be both faithful and constructible without circularity.

5.3. Parabolic backward uniqueness. The DBN flow is a (backward) heat flow, and backward uniqueness for parabolic equations would propagate $u(\Lambda) = 0 \Rightarrow u \equiv 0$. This works for quantities *linear* in Ξ_t . The reality-of-zeros detector is nonlinear (it sees the imaginary parts of zeros, not a linear functional of Ξ_t). This is the “linear $\neq$ nonlinear” gap identified as the terminus of Path III in [7]; backward uniqueness is available exactly where it is useless.

5.4. Monotonicity / integrality. The Pontryagin negative index $\kappa(Q) = \text{neg.ind}(H_C)$ is an integer ([5], with $\kappa = 2m$, localised in $\mathcal{K}_{\text{off}}$ by [7, Thm 8.1]) and upper semicontinuous. If a local principle forbade its jumps along a deformation to the on-line configuration, integrality would force $\kappa = 0$. But the jumps occur exactly at zero collisions, and nothing local prohibits them—generic heat flow *creates* complex pairs on crossing Λ for other entire functions. Integrality constrains the value but not its constancy.

5.5. Ergodic recurrence. Bagchi’s criterion makes RH equivalent to strong recurrence of the vertical flow $\tau \mapsto \zeta(\cdot + i\tau)$ on the Bohr torus $\Omega = \prod_p \mathbb{T}_p$. The Kronecker flow is uniquely ergodic, so the arithmetic point ω_0 is uniformly recurrent unconditionally; the obstruction is not recurrence but the continuity of the Euler observable $\Phi(\omega) = \prod_p (1 - \omega(p)p^{-s})^{-1}$ at the arithmetic point, which holds iff $\sum_p p^{-s}$ converges in the strip, iff RH [7, Prop. 7.2]. The ergodic route is MW-2 in dynamical language, with no residue.

5.6. Conclusion of the no-go.

Observation 5.2. Each classical 1D propagation mechanism, applied to the boundary-point families of §3, either coincides with RH (continuity §5.1, analyticity §5.2) or is excluded by a named wall (backward uniqueness by nonlinearity, integrality by collision jumps, ergodicity by MW-2). The 1D toolkit is exhausted.

Consequently, an argument that reaches RH cannot be a one-dimensional propagation principle. By elimination, the candidate logical forms that remain are: (A) two-dimensional propagation, where information flows across a region from its sides to a corner without passing continuously through any single boundary point; and (B) amplification, where a defect is multiplied until it collides with an unconditional sublinear bound. The rest of the paper specifies these.

6. FORM A: TWO-DIMENSIONAL PROPAGATION

6.1. Why a second dimension is not excluded. Every wall in §5 concerns a 1D boundary. MW-2 says the x -axis (arithmetic filtration) alone does not reach the corner; the DBN analysis says the t -axis (heat time) alone does not. *No result in the corpus addresses their coupling.* Two-dimensional propagation principles—the maximum principle, Almgren-type monotonicity, logarithmic convexity along diagonals—move information from the *edges* of a region into its interior and corners, and do not require continuity through any single point. This is the gap.

6.2. The candidate object. Define the joint trace

$$\mathcal{T}(t, x) = (\text{CCM trace of the finite-Euler approximant indexed by } x, \\ \text{evolved by the DBN heat flow to time } t).$$

Its edges and corner are controlled by results already in hand:

- Edge $\{t \geq \Lambda\}$: $\mathcal{T} = 0$ unconditionally (DBN, [7, Docs 70–71]).
- Edge $\{x \text{ finite}\}$: real spectrum unconditionally (CF rigidity of the CCM triples, [7,], which passed the circularity filter I2a at finite level).
- Corner $(t, x) = (0, \infty)$: RH.
- $\partial_t \mathcal{T}$ is computed: a quadratic form in the Jacobi data c_k , with exact cancellation under RH [7,].
- $\partial_x \mathcal{T}$ is computed in discrete form: primes enter one at a time through $T_\lambda = A_\lambda^{\text{off}} - \sum_p (\log p / \sqrt{p}) B_\lambda(\log p)$ [7,].

Problem 6.1 (The coupling inequality). Does $\mathcal{T}(t, x)$ satisfy a differential inequality coupling ∂_t and ∂_x —parabolic, monotone, or logarithmically convex along diagonals of the quadrant—under

which positivity propagates from the two edges to the corner $(0, \infty)$, even though it fails along each axis separately?

Remark 6.2 (What would falsify Form A). If $\partial_t \mathcal{T}$ and $\partial_x \mathcal{T}$ have no fixed sign relation on the quadrant—equivalently, if the mixed second variation $\partial_t \partial_x \mathcal{T}$ changes sign with no diagonal along which a monotone quantity is built—then no maximum/monotonicity principle applies and Form A is dead. The first test is finite and honest: compute $\partial_t \mathcal{T}$ for the two-prime approximant and compare its sign to $\partial_x \mathcal{T}$.

7. FORM B: INDEX AMPLIFICATION (THE DELIGNE MECHANISM)

7.1. The skeleton of the one proof of RH that exists. RH is a theorem over function fields [4]. Stripped to its logical skeleton, Deligne’s mechanism is not a positivity argument:

- (1) an *integer* invariant (the weight of a Frobenius eigenvalue);
- (2) an *amplification*: a tensor/symmetric power multiplies the defect, $\kappa \mapsto k\kappa$;
- (3) a *uniform sublinear bound* ($k\kappa \leq C(k)$ with $C(k)/k \rightarrow 0$, from rationality of the family’s L -function: the poles do not move);
- (4) conclusion: $k\kappa \leq C(k)$ for all k forces $\kappa = 0$.

None of (1)–(4) is a CAP-type positivity. The programme audited the *cohomological formalism* of Weil–Deligne (it is MW-5); it did not audit the *amplification trick* as an abstract skeleton. The skeleton is what we transplant.

7.2. What the programme already supplies. Ingredient (1) is in hand: $\kappa(Q) = \text{neg.ind}(H_C)$ is an integer, $\kappa = 2m$, localised in $\mathcal{K}_{\text{off}}$ ([5]; [7,]). The missing ingredient is the amplification.

Problem 7.1 (The amplifying functor). Does there exist an operation $\boxtimes$ on the Pontryagin space (K, Q) , or on the category of CCM spectral triples, such that

- (a) $\kappa(\psi_1 \boxtimes \psi_2) \geq \kappa(\psi_1) + \kappa(\psi_2)$ (superadditivity of the negative index), and
- (b) there is a bound $\kappa(\psi^{\boxtimes k}) \leq f(k)$ with $f(k)/k \rightarrow 0$, provable by a finiteness condition of type C5 ([7,])?

If both hold, then $2mk \leq f(k)$ for all k forces $m = 0$, i.e. RH.

7.3. Why it has not been tried, and where to look. For ζ —a $\text{GL}(1)$ object—the *grouplike* tensor theory is trivial: sym^k of the trivial character is trivial, which is why the Langlands route to Ramanujan-type bounds works for modular forms but says nothing about ζ . The operation $\boxtimes$ must therefore be *non-grouplike*: a candidate is the λ -ring / Frobenius-lift structure on the Connes–Consani scaling site, or a Rankin–Selberg-type convolution realised directly on (K, Q) rather than on automorphic data.

Remark 7.2 (What would falsify Form B). If κ is exactly additive under every available $\boxtimes$ (not strictly superadditive) and the bound $f(k)$ is genuinely linear, there is no contradiction to extract and Form B is dead. The decisive sub-question is whether *any* natural product on (K, Q) makes the negative index strictly superadditive.

Remark 7.3 (Resonance with the one non-CAP mechanism found). The single mechanism in the entire corpus that is provably not a CAP positivity is Bagchi’s amplification [7,]: one off-critical zero, replicated by strong recurrence to a positive-density family, contradicts the unconditional zero-density bound $N(\sigma, T) = o(T)$. Form B is that same skeleton with an algebraic amplifier in place of the ergodic one. The ergodic version failed only because its premise (strong recurrence) is RH-equivalent (§5); the skeleton itself is sound. This is the strongest single piece of evidence that the amplification *form* is the right place to look.

8. FORM C: THE BOUNDARY POINT AS A QUANTITATIVE SINGULARITY

A cheaper, lower-risk line. The self-approximation family SR_d is proved unconditionally for all $d \neq 0$, with $d = 0 \iff \text{RH}$ [7,]. The effective question has not been asked.

Problem 8.1 (Effective degeneration of SR_d). As $d \rightarrow 0$, how do the constants in the proof of SR_d degrade? If the degradation is $e^{-c/|d|}$, this is quantified MW-2 and the line closes cleanly. If it is polynomial in $1/|d|$, there is a genuine tension with the Rouché argument linking recurrence to zero density, deserving a dedicated analysis.

Form C produces knowledge either way: a closed line (with MW-2 quantified) or a tension to exploit. It is the natural first move because it is finite, falsifiable, and independent of the harder Forms A and B.

9. HONEST ASSESSMENT

We claim no proof and no partial proof of RH. We claim three things. First, a structural no-go (Observation 5.2): the one-dimensional propagation toolkit is exhausted against the boundary-point families that organise the subject, and we have checked the mechanisms individually. Second, one new unconditional theorem internal to the framework (Theorem 4.4): the trace criterion holds at a single scale, the λ -family being spurious. Third, a forward agenda (Problems 6.1, 7.1, 8.1) that is precisely the set of logical forms the no-go does not exclude, each with a stated falsifier.

If Forms A and B also collapse onto known walls, that collapse will itself be informative: it would show the obstruction is deeper than the 1D/2D/amplification trichotomy can express, and would be the first evidence that RH is unreachable by *any* of the argument-forms currently known to mathematics. That, too, would be a result. The honest position is that we have narrowed the search for the missing idea from “anything” to “a non-grouplike amplification of the Pontryagin index, or a coupled monotonicity on the heat-time/filtration quadrant,” and that this narrowing—not any claim of progress toward the proof itself—is the contribution.

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